

Juniper Automated Test Station

A Python/Flask application driving the Juniper test bench: the **Vitrek V71 Hi-Pot Tester** over USB or RS-232, a **Siglent SDL1020X-E** DC electronic load, and an always-on **Siemens LOGO! PLC** thermal rig. Results are stored in SQLite and exported to a branded Excel workbook for SharePoint. Built at [Juniper Design](#).

 **Print-ready PDF:** `docs/pdf/README.pdf`

Features

- **Two access levels** — operators run approved sequences with no login; admins define them. See `docs/access-control.md`
- **Saved test sequences** — reviewed, versioned definitions rather than parameters typed at the bench
- **Instrument verification (PVD)** — daily performance verification against a Vitrek APVD, with the actual measured values recorded, not just a pass light. See `docs/pvd-verification.md`
- **USB and RS-232** — USB via the Silicon Labs CP2110 HID-to-UART DLL; RS-232 via pyserial
- **Full V7X command set** — ACW, DCW, IR, GB, CONT, PAUSE, HOLD steps
- **DC load + thermal rig** — SDL1020X-E load batteries, PEC-0063 thermal qualification, continuous 1 Hz sensor recording
- **Web UI** — nothing to install on operator machines, just open the page
- **Excel export** — colour-coded workbook covering test sessions, thermal qualification and instrument verification

Quick Start

```
pip install -r requirements.txt
python app.py
```

Open `http://localhost:5000`.

Connect the V71: set `CONFIG MENU → INTERFACE = USB` on the front panel and plug in the USB-B cable, or `= RS232` with a fully-wired 9-wire null-modem cable (hardware RTS/CTS handshaking is required).

First run

1. Click **Set up admin** in the top bar and choose a password. It is stored only as a PBKDF2 hash — keep the password itself in 1Password.
 2. As admin, open **HiPot** → **Sequence Builder** and create at least one sequence. Operators cannot run anything until one exists.
 3. Baseline the PVD profile before relying on verification verdicts — see `docs/pvd-verification.md`.
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Instruments

Instrument	Interface	Notes
Vitrek V71 HiPot	USB HID-to-UART, or RS-232	ACW, DCW and CONT modes only — no IR, no GB
Siglent SDLI020X-E DC Load	TCP/LAN, VISA or USB CDC	
Siemens LOGO! PLC + thermal rig	Modbus TCP	Always on, independent of instrument selection

Instrument selection is mutually exclusive — one at a time — but the thermal rig runs in parallel with any instrument test, and sensor data is continuously recorded so results can be correlated against the timeline.

Project Structure

```
├─ app.py           Flask app: REST API + embedded UI for every page
├─ auth.py          Roles, admin password hashing, @admin_required
├─ v71_driver.py    V71 USB/serial driver (ctypes + pyserial)
├─ sd11020x_driver.py Siglent DC load driver
├─ pvd_test.py       Instrument verification runner
├─ pvd_profiles.json Verification profiles (APVD-74 / APVD-7X)
├─ test_battery.py   Multi-step DC load test batteries
├─ pec0063_test.py   PEC-0063 thermal qualification
├─ database.py       SQLite schema and CRUD
├─ excel_export.py   Branded Excel workbook generator
├─ plc/             LOGO! PLC driver, thermal controller, rig_config.json
├─ hipot_results.db  SQLite database (auto-created; gitignored)
├─ docs/
│   ├─ access-control.md
│   ├─ pvd-verification.md
│   ├─ wiring-guide.md
│   ├─ hmi-architecture.md
│   ├─ plc-ladder-logic.md
│   ├─ breadboard-prototype.md
│   ├─ sourcing-list.md
│   └─ V7x_Series_Operating_Manual.pdf
```

REST API

Routes marked **admin** require an admin session; everything else is available to operators. Full detail in the two docs above.

Auth and setup

Method	Endpoint		Description
GET	/api/auth/status		Current role
POST	/api/auth/login		Elevate to admin
POST	/api/auth/logout		Drop to operator
POST	/api/auth/set_password		Set or change the admin password
GET/POST	/api/settings/connection	admin to write	Saved connection settings

Test sequences

Method	Endpoint		Description
GET	/api/sequences		List saved sequences
POST	/api/sequences	admin	Create
PUT	/api/sequences/<id>	admin	Update (bumps revision)
DELETE	/api/sequences/<id>	admin	Retire (?hard=1 deletes)

HiPot

Method	Endpoint		Description
POST	/api/connect		Connect (operators use the saved settings)
POST	/api/disconnect		Disconnect
POST	/api/hipot/run_sequence		Run a saved sequence
POST	/api/hipot/run	admin	Run ad-hoc steps
POST	/api/hipot/abort · /api/hipot/cont		Abort / continue
GET	/api/hipot/status · /api/hipot/live		Run state and live V/A/Ω

Verification, results and export

Method	Endpoint		Description
POST	/api/pvd/start · /api/pvd/ack · /api/pvd/stop	admin	Run a verification
POST	/api/pvd/baseline/promote	admin	Adopt baseline measurements as nominals
GET	/api/pvd/results · /api/pvd/verification_status		Verification history and currency
GET	/api/sessions · /api/session/<id>		Test history
GET	/api/export · /api/export/<id>		Excel export

USB Communication Details

The V71 uses a **Silicon Labs CP2110 HID-to-UART bridge**:

- SLABHIDtoUART.dll + SLABHIDDevice.dll (x64, from software/drivers/)
- USB VID 4292 (0x10C4), PID 34869 (0x8835)

- 115200 baud, 8N1, RTS/CTS flow control
- ASCII commands terminated with `\r\n`

Key commands (Section 6 of the operating manual):

Command	Description
<code>*IDN?</code> / <code>*RST</code> / <code>*ERR?</code>	Identify / reset / read error register
<code>NOSEQ</code>	Clear and activate sequence #0
<code>ADD,ACW,...</code> <code>ADD,DCW,...</code> <code>ADD,IR,...</code> <code>ADD,GB,...</code> <code>ADD,CONT,...</code>	Add a test step
<code>ADD,PAUSE,...</code> <code>ADD,HOLD,...</code>	Timed pause / operator prompt
<code>RUN</code> · <code>ABORT</code> · <code>CONT</code> · <code>RUN?</code>	Execution control
<code>RSLT?</code> · <code>STAT?</code> · <code>STEPRSLT?,<n></code>	Overall, per-step and detailed results
<code>MEASRSLT?,<AMPS\ VOLTS\ OHMS></code>	Live measurement

There is **no remote command for AUTO PVD, SELF TEST or CAL VERIFY** — those are front-panel only, which is why verification is implemented as a programmed sequence. See `docs/pvd-verification.md`.

Excel Export

One workbook, several sheets:

- **Summary** — all test sessions, sortable and filterable
- **Per-session** — full step results, colour-coded
- **PEC-0063 Thermal Results** — thermal qualification against the UL limits
- **PVD Verification** — instrument verification, one row per measured point

Suitable for direct upload or auto-sync to SharePoint.

Packaging & deployment

The bench machine (**HI-POT-TEST**) does not run this from source — it runs a packaged single-file executable, deployed and kept current by the Juniper inventory server.

- **Build the exe:** `powershell -ExecutionPolicy Bypass -File build-exe.ps1 -Version <x.y.z>`. This produces `dist/HiPotController-<x.y.z>.exe` (PyInstaller **onefile**, windowed) and prints its SHA256 + size. The build bundles `static/`, `pvd_profiles.json`, `plc/rig_config.json`, and the x64 Silicon Labs CP2110 DLLs at the path `v71_driver.py` loads them from. `build/` and `dist/` are gitignored — the exe is a build artefact, published to the server, not committed.
- **Persistent data (important for packaging).** Under a onefile exe, `os.path.dirname(__file__)` is the temporary extraction dir that Windows deletes on exit, so the app writes its **results DB**

and the **mutable** `pvd_profiles.json` (baseline promotion rewrites it) to `C:\ProgramData\Juniper\HiPotController` instead — see `paths.py` (`data_dir()` / `bundled_dir()`). Read-only bundled resources (`rig_config.json`, brand assets) are read from the bundle. Running from source is unchanged: `data_dir()` resolves to the repo, so dev keeps its files in place.

- **Deploying to the fleet:** the exe is uploaded to the inventory server and the catalog package "**Juniper HiPot Controller**" (device-scoped to HI-POT-TEST) is bumped to the new version, with its install script pinned to the new exe's hash. The agent installs to `C:\Program Files\Juniper\HiPot Controller` on next check-in.

Notes

- The DLLs must be present at `software/drivers/USB_DLLs_and_Headers/USB DLLs and Headers/x64/`, and the app must run as **x64** to match them
- After a DUT breakdown on USB the V71 may disconnect and reconnect; the driver surfaces the error — just reconnect from the UI
- RS-232 requires hardware handshaking; a 3-wire cable will not work
- `hipot_results.db` is gitignored: it holds the admin password verifier and the session signing key, neither of which belongs in the repository